

University Institute of Engineering

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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User Interface Design

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CSE

DISCOVER . **LEARN** . EMPOWER

Chapter-4

User Interface Design

- Command language user interface
- Menu based ,Direct manipulation interfaces

Types of User Interface

UI is broadly divided into the following categories:

- Command Line Interface
- Graphical User Interface
- Menu based UI
- Direct Manipulation Interface

Command Line Interface (CLI)

- A **Command Line Interface (CLI)** is a type of user interface where users interact with a computer or software by typing text-based commands. Unlike graphical user interfaces (GUIs), CLIs rely on keyboard input and do not use icons, buttons, or windows.

Examples of CLI Applications

1. Operating System Shells

1. Windows Command Prompt (`cmd.exe`)
2. PowerShell (for advanced Windows scripting)
3. Linux Terminal (`bash`, `zsh`, `fish`)
4. macOS Terminal (uses Unix-based shell like `zsh` or `bash`)

2. Programming and Development

1. Git (`git commit -m "message"`) for version control.
2. Python, Node.js, Java compilers (e.g., `python script.py`, `javac HelloWorld.java`).

3. Database Management

1. MySQL, PostgreSQL, MongoDB (e.g., `mysql -u root -p`, `psql -d mydatabase`).

4. Networking and System Administration

1. Ping and Traceroute (`ping google.com`, `tracert 8.8.8.8`).
2. SSH for remote access (`ssh user@server.com`).
3. File operations (`mv`, `cp`, `rm`, `mkdir`).

Advantages of CLI

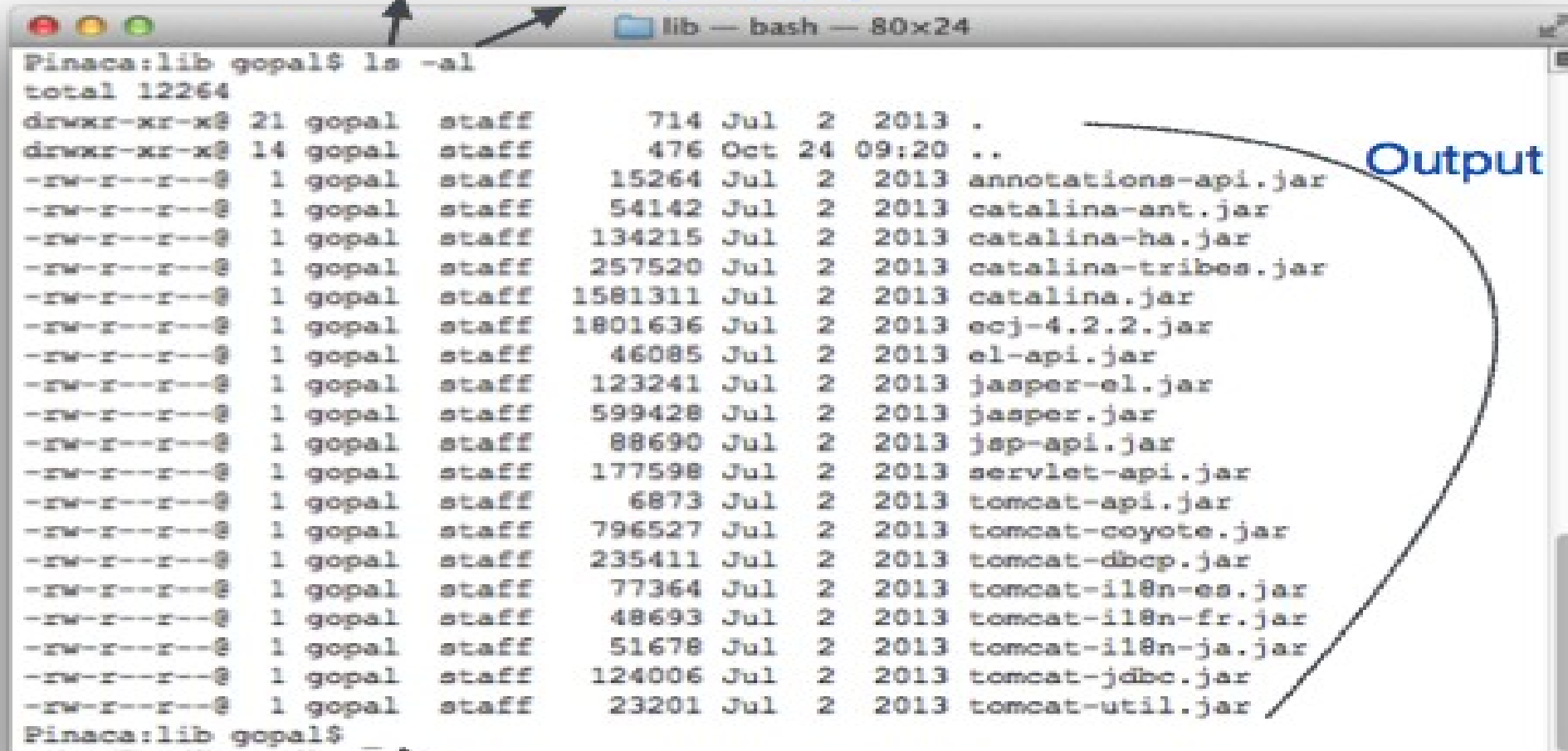
- ✓ **Efficient for power users** – Faster than navigating a GUI.
- ✓ **Low resource consumption** – Runs on minimal hardware.
- ✓ **Highly flexible** – Supports scripting and automation.
- ✓ **Remote Access** – Used for managing servers and networks.

Disadvantages of CLI

- ✗ **Steep learning curve** – Requires memorization of commands.
- ✗ **Error-prone** – A single typo can cause unintended actions.
- ✗ **Not user-friendly** – Difficult for beginners compared to GUI.

CLI Elements

Command Parameters



```
Pinaca:lib gopal$ ls -al
total 12264
drwxr-xr-x 21 gopal staff 714 Jul 2 2013 .
drwxr-xr-x 14 gopal staff 476 Oct 24 09:20 ..
-rw-r--r-- 1 gopal staff 15264 Jul 2 2013 annotations-api.jar
-rw-r--r-- 1 gopal staff 54142 Jul 2 2013 catalina-ant.jar
-rw-r--r-- 1 gopal staff 134215 Jul 2 2013 catalina-ha.jar
-rw-r--r-- 1 gopal staff 257520 Jul 2 2013 catalina-tribes.jar
-rw-r--r-- 1 gopal staff 1581311 Jul 2 2013 catalina.jar
-rw-r--r-- 1 gopal staff 1801636 Jul 2 2013 ecj-4.2.2.jar
-rw-r--r-- 1 gopal staff 46085 Jul 2 2013 el-api.jar
-rw-r--r-- 1 gopal staff 123241 Jul 2 2013 jasper-el.jar
-rw-r--r-- 1 gopal staff 599428 Jul 2 2013 jasper.jar
-rw-r--r-- 1 gopal staff 88690 Jul 2 2013 jsp-api.jar
-rw-r--r-- 1 gopal staff 177598 Jul 2 2013 servlet-api.jar
-rw-r--r-- 1 gopal staff 6873 Jul 2 2013 tomcat-api.jar
-rw-r--r-- 1 gopal staff 796527 Jul 2 2013 tomcat-coyote.jar
-rw-r--r-- 1 gopal staff 235411 Jul 2 2013 tomcat-dbcp.jar
-rw-r--r-- 1 gopal staff 77364 Jul 2 2013 tomcat-el8n-es.jar
-rw-r--r-- 1 gopal staff 48693 Jul 2 2013 tomcat-el8n-fr.jar
-rw-r--r-- 1 gopal staff 51678 Jul 2 2013 tomcat-el8n-ja.jar
-rw-r--r-- 1 gopal staff 124006 Jul 2 2013 tomcat-jdbc.jar
-rw-r--r-- 1 gopal staff 23201 Jul 2 2013 tomcat-util.jar
Pinaca:lib gopal$
```

Output

Cursor

Command Prompt

CLI Elements

A text-based command line interface can have the following elements:

- **Command Prompt** - It is text-based notifier that is mostly shows the context in which the user is working. It is generated by the software system.
- **Cursor** - It is a small horizontal line or a vertical bar of the height of line, to represent position of character while typing. Cursor is mostly found in blinking state. It moves as the user writes or deletes something.
- **Command** - A command is an executable instruction. It may have one or more parameters. Output on command execution is shown inline on the screen. When output is produced, command prompt is displayed on the next line.

Graphical User Interface

- A **Graphical User Interface (GUI)** is a type of user interface that allows users to interact with a system using visual elements such as windows, icons, buttons, and menus instead of text-based commands. It is designed to be intuitive and user-friendly, making computers accessible to people with little technical knowledge.

Examples of GUI-Based Systems

1. Operating Systems

1. **Microsoft Windows** (Start menu, taskbar, icons)
2. **macOS** (Dock, Finder, menu bar)
3. **Linux (Ubuntu, Fedora, etc.)** (GNOME, KDE environments)

2. Software Applications

1. **Microsoft Office Suite** (Word, Excel, PowerPoint)
2. **Adobe Photoshop** (Toolbars, color palettes)
3. **Web Browsers** (Google Chrome, Mozilla Firefox, Safari)

3. Mobile Operating Systems

1. **Android** (App icons, notification bar)
2. **iOS** (Home screen, control center, gestures)

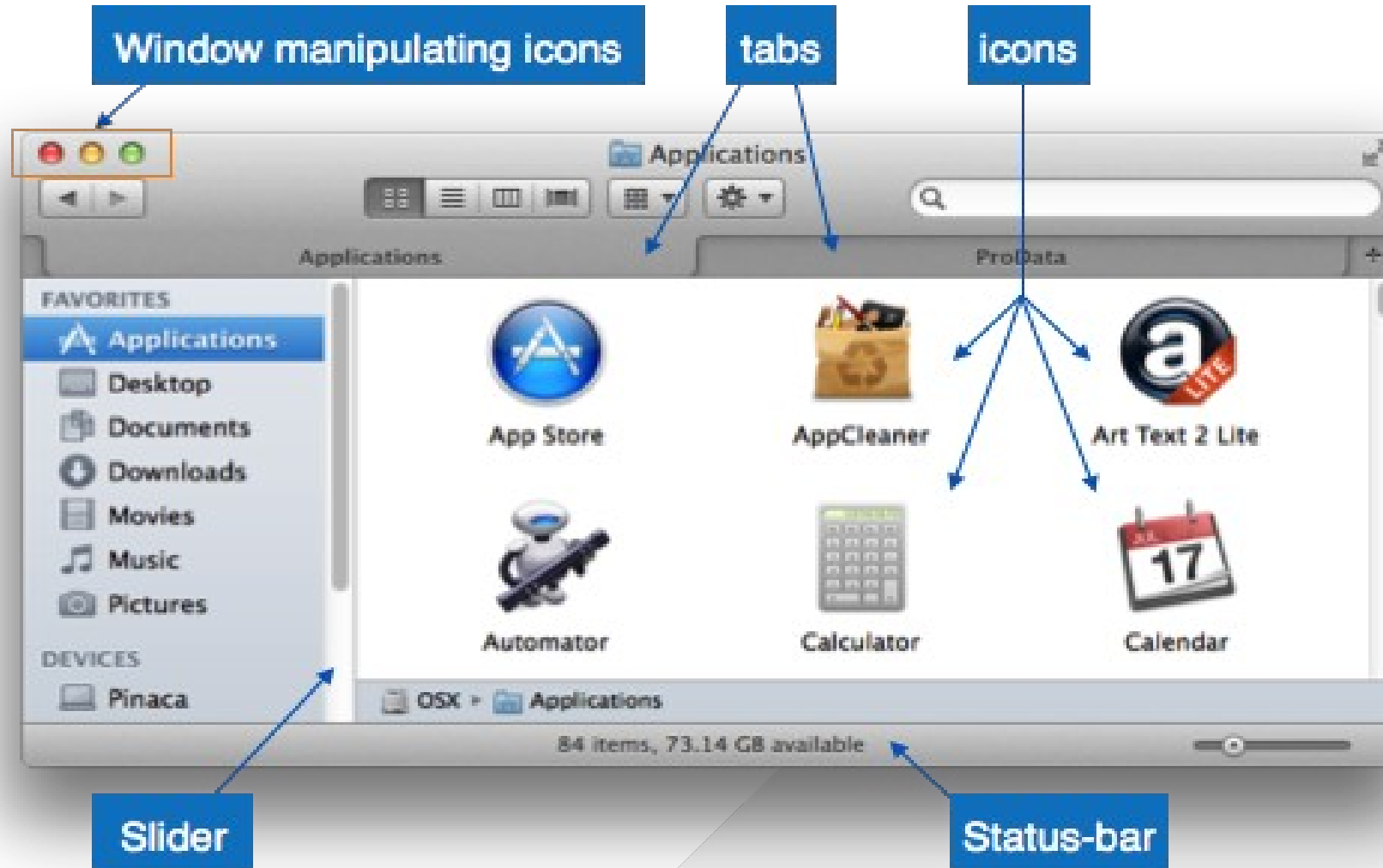
Advantages of GUI

- ✓ **User-Friendly** – Easy to learn and use, even for beginners.
- ✓ **Visual and Interactive** – Enhances user experience with graphics and animations.
- ✓ **Multitasking** – Allows users to open and manage multiple applications simultaneously.
- ✓ **Minimal Memorization** – No need to remember complex commands like in CLI.

Disadvantages of GUI

- ✗ **Resource-Intensive** – Requires more memory and processing power compared to CLI.
- ✗ **Slower for Advanced Users** – Some tasks can be done faster using command-line interfaces.
- ✗ **Limited Customization** – Not as flexible as scripting in CLI for automation and complex tasks.

GUI Elements



GUI Elements

- **Tabs** - If an application allows executing multiple instances of itself, they appear on the screen as separate windows. **Tabbed Document Interface** has come up to open multiple documents in the same window. This interface also helps in viewing preference panel in application. All modern web-browsers use this feature.
- **Menu** - Menu is an array of standard commands, grouped together and placed at a visible place (usually top) inside the application window. The menu can be programmed to appear or hide on mouse clicks.

GUI Elements

- **Icon** - An icon is small picture representing an associated application. When these icons are clicked or double clicked, the application window is opened. Icon displays application and programs installed on a system in the form of small pictures.
- **Cursor** - Interacting devices such as mouse, touch pad, digital pen are represented in GUI as cursors. On screen cursor follows the instructions from hardware in almost real-time. Cursors are also named pointers in GUI systems. They are used to select menus, windows and other application features.

- **Text-Box** - Provides an area for user to type and enter text-based data.
- **Buttons** - They imitate real life buttons and are used to submit inputs to the software.
- **Radio-button** - Displays available options for selection. Only one can be selected among all offered.
- **Check-box** - Functions similar to list-box. When an option is selected, the box is marked as checked. Multiple options represented by check boxes can be selected.
- **List-box** - Provides list of available items for selection. More than one item can be selected.

Menu-Driven Interface

- The menu-driven interface employs a series of screens, or “menus”. When a user makes a selection by tapping/clicking on the list format or graphics, it takes them to the next menu screen until they complete the desired outcome.
- An example is the settings menu on your phone. All you can do is scroll the menu and tap items; no other interaction is available. Accordingly, it is used in applications with a familiar, limited, and uniform set of functions. You could put them all under subheadings like “Camera Settings”, “Display Settings”, etc.

Examples of Menu-Based Interfaces

1.ATMs (Automated Teller Machines)

1. Users navigate options such as "Withdraw Cash," "Check Balance," or "Deposit Funds" using on-screen buttons or a touchscreen.

2.Restaurant Self-Ordering Kiosks

1. Customers select food and drinks from a digital menu displayed on a touchscreen.

3.Mobile Phone Settings

1. Smartphones have menu-based interfaces for settings like "Wi-Fi," "Bluetooth," and "Display" options.

4.Video Game Menus

1. Games have menus for selecting "Start Game," "Load Game," "Settings," and "Exit."

5.Software Applications (e.g., Microsoft Word, Excel)

1. Features like "File," "Edit," "View," and "Tools" are organized in a menu for easy access.

Advantages of a Menu-Driven Interface

- Handy for computer beginners and novice users
- Familiar interface across different platforms
- You are in charge of creating an order and a hierarchy for user pathways
- More control over user interactions
- Simple to implement in various kinds of devices

Disadvantages of a Menu-Driven Interface

- Limited menu options
- Sub-menus might be difficult to find
- Risks of taking up a lot of screen space or being too small
- Requires unnecessary actions for a simple task

Direct Manipulation Interface

A **Direct Manipulation Interface (DMI)** in software engineering is a type of user interface that allows users to interact directly with digital objects rather than relying on complex command-line inputs or abstract interactions.

Examples of Direct Manipulation Interfaces

- **Graphical Operating Systems** (e.g., Windows, macOS)
 - Dragging and dropping files, resizing windows.
- **Photo Editing Software** (e.g., Photoshop)
 - Adjusting image properties by dragging sliders.
- **Touchscreen Interfaces** (e.g., smartphones, tablets)
 - Pinching to zoom, swiping to navigate.
- **Game Interfaces**
 - Controlling characters using joysticks or on-screen buttons.

Advantages of Direct Manipulation Interfaces

- ✓ **User-friendly** – Easy for beginners to use.
- ✓ **Efficient** – Faster than typing commands.
- ✓ **Encourages Exploration** – Users can try different actions with immediate feedback.
- ✓ **Reduces Errors** – The ability to undo mistakes makes the system more robust.

Disadvantages of Direct Manipulation Interfaces

- ✗ **Resource-Intensive** – Requires more processing power and memory than text-based interfaces.
- ✗ **Not Ideal for Complex Tasks** – Some advanced operations are harder to perform without command-line precision.
- ✗ **Scalability Issues** – Can become cluttered when dealing with large datasets or complex workflows.

References

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- <https://www.elegantthemes.com/blog/resources/10-rules-of-good-ui-design-to-follow-on-every-web-design-project>



THANK YOU